

ML Specification Reports — 23 Competitions (the 20-competition benchmark baseline + 3 special deep-learning-fallback lanes)

Each competition is documented as a **report** against the 顏佐榕 ML-spec framework, satisfying the summer-project plan's Goal 1: (甲) 競賽目的 — **what + why** (Overview + "Why it matters") and (乙) 五大流程元件 — Data / Models / Training / Inference / Evaluation. Presentation is narrative-first with tables only where genuinely tabular; every figure is grounded in that competition's `facts.json` / `config.yaml` / `eda_summary.json` / `STATUS.md` / `experiments_tree_v3.json`, and DL-only items with no tabular-GBDT analogue are marked N/A (reason).

Each report has a Markdown source (`<comp>.md`) and an individual academic-serif PDF (`../pdf/<comp>.pdf`). Rubric validation: `RUBRIC.md`. Rows 10 (s3e20) and 11 (s4e1) below document competitions **outside** the frozen baseline; their reports were moved to `../archive_non_benchmark/`, so this folder holds **23** reports — the 20 baseline competitions plus the 3 special lanes tabulated at the end.

#	Competition	Task	Metric	Our best (CV)	NVIDIA	Winner
1	s3e1	California Housing — median house value	RMSE ↓	0.5571	0.5086	NVIDIA
2	s3e3	Employee attrition	AUC ↑	0.845051	0.8758	NVIDIA
3	s3e5	Wine quality (ordinal)	QWK ↑	0.5677	0.5478	our agent
4	s3e7	Hotel reservation cancellation	AUC ↑	0.8999	0.9014	tie
5	s3e9	Concrete compressive strength	RMSE ↓	12.07	12.026	tie
6	s3e11	Media campaign cost	RMSLE ↓	0.29565	0.2956	tie
7	s3e14	Wild blueberry yield	MAE ↓	340.6	337.28	NVIDIA
8	s3e16	Crab age	MAE ↓	1.3356	1.3387	our agent
9	s3e19	Forecast mini-course sales	SMAPE ↓	10.02	13.72	our agent
10	s3e20	Rwanda CO2 emission	RMSE ↓	21.15	19.648	NVIDIA
11	s4e1	Bank customer churn	AUC ↑	0.894393	0.8984	NVIDIA
12	s4e11	Depression prediction	Accuracy ↑	0.9402	0.9399	tie
13	s5e10	Road accident risk	RMSE ↓	0.05597	0.05609	tie

#	Competition	Task	Metric	Our best (CV)	NVIDIA	Winner
14	s6e1	Student exam-score	R ² ↑	0.78718	0.7863	tie
15	s6e2	Heart disease	AUC ↑	0.95553	0.9554	tie

Record vs NVIDIA reproduce-agent: **3 our-agent wins / 7 ties / 5 NVIDIA wins**. Scores are local CV (all competitions closed → no live leaderboard); benchmark methodology and caveats (de-leaking, unified CV schemes) are documented per report and in the four-axis evaluation.

Do not mix this CV-based record with the midterm report's LB-based standings. The midterm report (`.../MIDTERM_REPORT_202607.md`) scores the same competitions by late-submission leaderboard results with a paired significance test; this README's table is local CV. The two disagree in 10 of 13 comparable competitions (s3e19 fully reverses: CV says our agent wins by 27%, private LB says NVIDIA wins), because local CV schemes differ per agent and can leak (see the midterm report § 3.4). Cite LB numbers for any between-agent claim; CV numbers here describe each pipeline's internal view only.

Benchmark-baseline additions (added 2026-08-17)

The seven competitions below complete coverage of the frozen 20-competition benchmark baseline (REPORT_v3/v4 § The Three-Way Benchmark). Unlike the CV-based table above, these rows quote the **real private leaderboard** from the study's frozen three-way score table, per the rule stated in the note — LB numbers are the ones fit for between-agent claims. PR = percentile rank.

#	Competition	Metric	our private LB (PR)	NVIDIA	AIDE	LB winner
16	afsis-soil-properties	MCRMSE ↓	0.49517 (46.6)	0.69855	0.49861	our agent
17	cat-in-the-dat	ROC-AUC ↑	0.80241 (73.1)	0.77084	0.80217	our agent
18	conway-s-reverse-game-of-life	MAE ↓	0.10875 (98.6)	0.11189	0.11065	our agent
19	tps-aug-2022	ROC-AUC ↑	0.59055 (66.0)	0.58730	0.59106	AIDE
20	tps-jan-2022	SMAPE ↓	6.46156 (71.5)	4.63651	10.66638	NVIDIA
21	s5e1	MAPE ↓	0.12752 (76.5)	0.14594	0.17917	our agent
22	tps-sep-2022	SMAPE ↓	28.54859 (14.6)	5.48351	9.72095	NVIDIA

s5e1 and tps-sep-2022 ran the full v5 pipeline (dossier + external-data injection, knowledge library pinned) and joined the frozen baseline on 2026-07-31; the other five ran in the 9a coverage phase.

Final-architecture re-run (Stage 0.5 in every lane). Each of the 20 baseline reports ends with a "Final-Architecture Re-Run" section: those competitions were re-run in an isolated root under one frozen v5 architecture where **each lane runs the Stage 0.5 problem dossier in-lane**, holds no credentials, and passes a per-lane transcript audit. All 20 lanes ran and were scored by the credentialed operator step; the public/private scores and `winner_priv` for each are recorded in `benchmark_results/rerun_three_way.csv` and quoted in that report's re-run section. Nothing in the historical sections above

is revised by these entries. s3e20 and s4e1 sit outside the baseline and have no re-run lane, and the three special lanes below are graded offline and have no re-run lane either. tps-sep-2022 is our agent's known failure mode (wrong country-level regime bet, flagged as undecidable-from-data before submission — see that report's Evaluation section).

Special deep-learning-fallback lanes (offline MLE-bench grading)

The three competitions below are the special lanes counted in this folder's 23 reports. They are graded **offline by MLE-bench** against a held-out private split — there is no live Kaggle leaderboard, so the figures are grades, not LB scores, and they are not comparable to the two tables above. Values copied from each report's Evaluation & Benchmarking table.

#	Competition	Metric	our grade	NVIDIA	AIDE	Winner	Medal (ours)
23	ventilator-pressure-prediction	masked MAE ↓	0.16525	0.40093	0.34591	our agent	none (median 0.1638)
24	us-patent-phrase-to-phrase-matching	Pearson r ↑	0.86658	0.8616	0.65821	our agent	silver (threshold 0.863)
25	siim-isic-melanoma-classification	ROC-AUC ↑	0.93529	0.89822	0.88579	our agent	none (bronze 0.937, median 0.9128)